CS-1204
Object-Oriented Programming
(3+1)

Lecture

By

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File Handling in C++

Objectives

- Understand the need for file handling.
- Learn about C++ file handling classes.
- Explore file modes.
- Understand file pointers and their usage.
- Practice with examples to strengthen concepts.

Key Points

- Variables store data in temporary memory (RAM).
- After program ends → data is lost.
- To keep data permanently → we use files stored on disk.

Classes for File Handling

- C++ provides three main classes
  - **ifstream** → Input file stream (for reading).
  - **ofstream** → Output file stream (for writing).
  - **fstream** → File stream (for both reading & writing).

```
#include <fstream>
using namespace std;

ifstream fin;   // read
ofstream fout;  // write
fstream file;   // read + write
```

Must include *fstream* header file to use input and output file stream.

Steps to use a file

- Declare a stream object (ifstream, ofstream, fstream).
- Open the file using constructor.
- Perform read/write operations.
- Close the file with close().

```
ofstream fout("data.txt"); // open file for writing
fout << "Hello File";      // write data
fout.close();              // close file
```

File open modes in C++:

- ios::out → Open file for writing.
- ios::in → Open file for reading.
- ios::app → Append data at the end.
- ios::trunc → Delete previous content.

Combining file modes

- Modes can be combined using |

```cpp
fstream file("test.txt", ios::out | ios::in);
```

- Open file for **both reading & writing**.

```cpp
ofstream fout("log.txt", ios::app | ios::out);
```

- Open file in **append mode** (previous data preserved).

Open file for writing.

source.cpp iostream

```cpp
#include<iostream>
#include<fstream>
using namespace std;
int main(){
    ofstream out("greet.txt");
    out << "Welcome to the File Handling Lecture";
    out.close();
    return 0;
}
```

source.cpp iostream

```cpp
#include<iostream>
#include<fstream>
using namespace std;
int main(){
    ofstream out("greet.txt", ios::out);
    out << "Welcome to the File Handling Lecture";
    out.close();
    return 0;
}
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source.cpp iostream

```cpp
#include<iostream>
#include<fstream>
using namespace std;
int main(){
    fstream out("greet.txt", ios::out);
    out << "Welcome to the File Handling Lecture";
    out.close();
    return 0;
}
```

source.cpp iostream

```cpp
#include<iostream>
#include<fstream>
using namespace std;
int main(){
    fstream out("greet.txt", ios::out | ios::trunc);
    out << "Welcome to the File Handling Lecture";
    out.close();
    return 0;
}
```

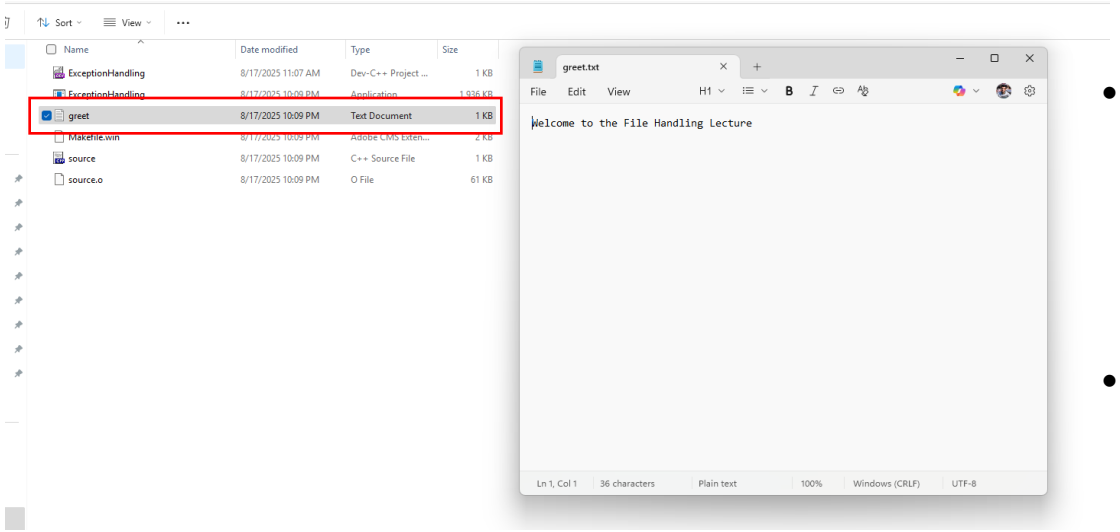
*] source.cpp iostream

```cpp
#include<iostream>
#include<fstream>
using namespace std;
int main(){
    fstream out("greet.txt", ios::out | ios::app);
    out << "Welcome to the File Handling Lecture";
    out.close();
    return 0;
}
```

ios::out → Open file for writing.

source.cpp iostream

```
1  #include<iostream>
2  #include<fstream>
3  using namespace std;
4  int main(){
5      fstream out("greet.txt", ios::out);
6      out << "Welcome to the File Handling Lecture";
7      out.close();
8      return 0;
9  }
```

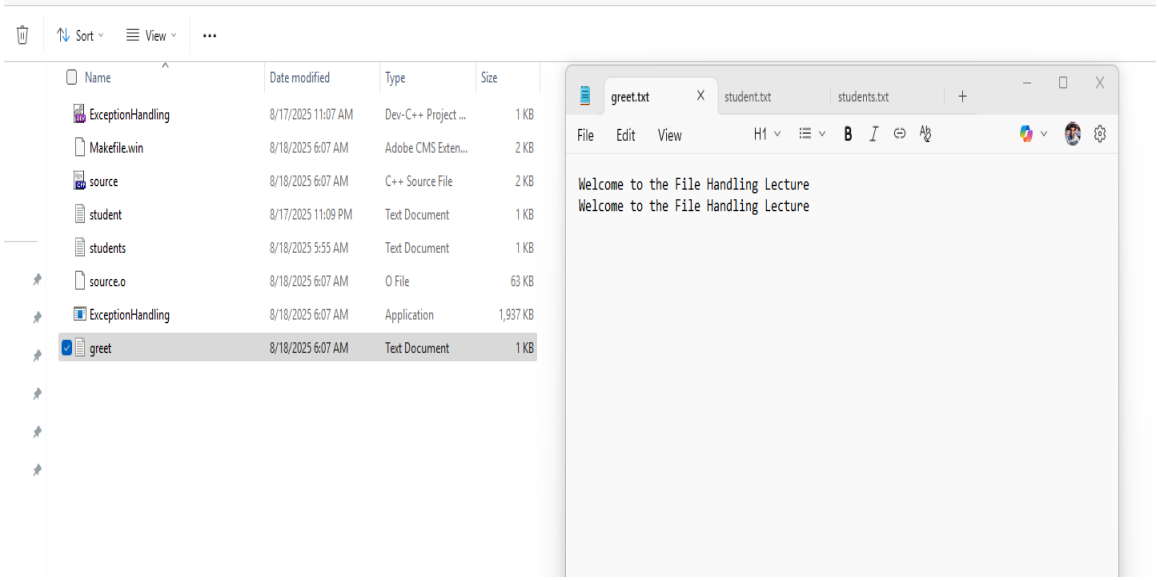


Opening a file in write mode (ios::out) will:

- Create the file if it doesn't already exist.
- Truncate (delete old content) if the file already exists.
- By default, when you open a file without specifying a truncate or append mode it opens in truncate mode.
- Each time you run the program, the file content will be overridden (old content is erased).

ios::app → Open file for writing in Append Mode.

```cpp
#include<iostream>
#include<fstream>
using namespace std;
int main(){
    fstream out("greet.txt", ios::out | ios::app);
    out << "Welcome to the File Handling Lecture" << endl;
    out.close();
    return 0;
}
```



Opening a file in append mode with (ios::app) will:

- Create the file if it doesn't already exist.
- Start writing from the end of the old content.
- The result on the left shows that the lines are appended at end when we run the program twice.

Writing to a File

```
#include <fstream>
#include <iostream>
using namespace std;

int main() {
    ofstream fout("student.txt");
    fout << "Roll No: 101" << endl;
    fout << "Name: Ali" << endl;
    fout.close();
    cout << "Data written successfully!";
}
```

Similar to, how we use *cout* for writing on a standard console. Here we use file stream object *fout* to write in the files.

Reading from a File

```
#include <fstream>
#include <iostream>
using namespace std;

int main() {
    ifstream fin("student.txt");
    string line;
    while (getline(fin, line)) {
        cout << line << endl;
    }
    fin.close();
}
```

Similar to, how we use *cin* for taking input from the user.
Here we use file stream object *fin* to read line from the file.

File Pointers

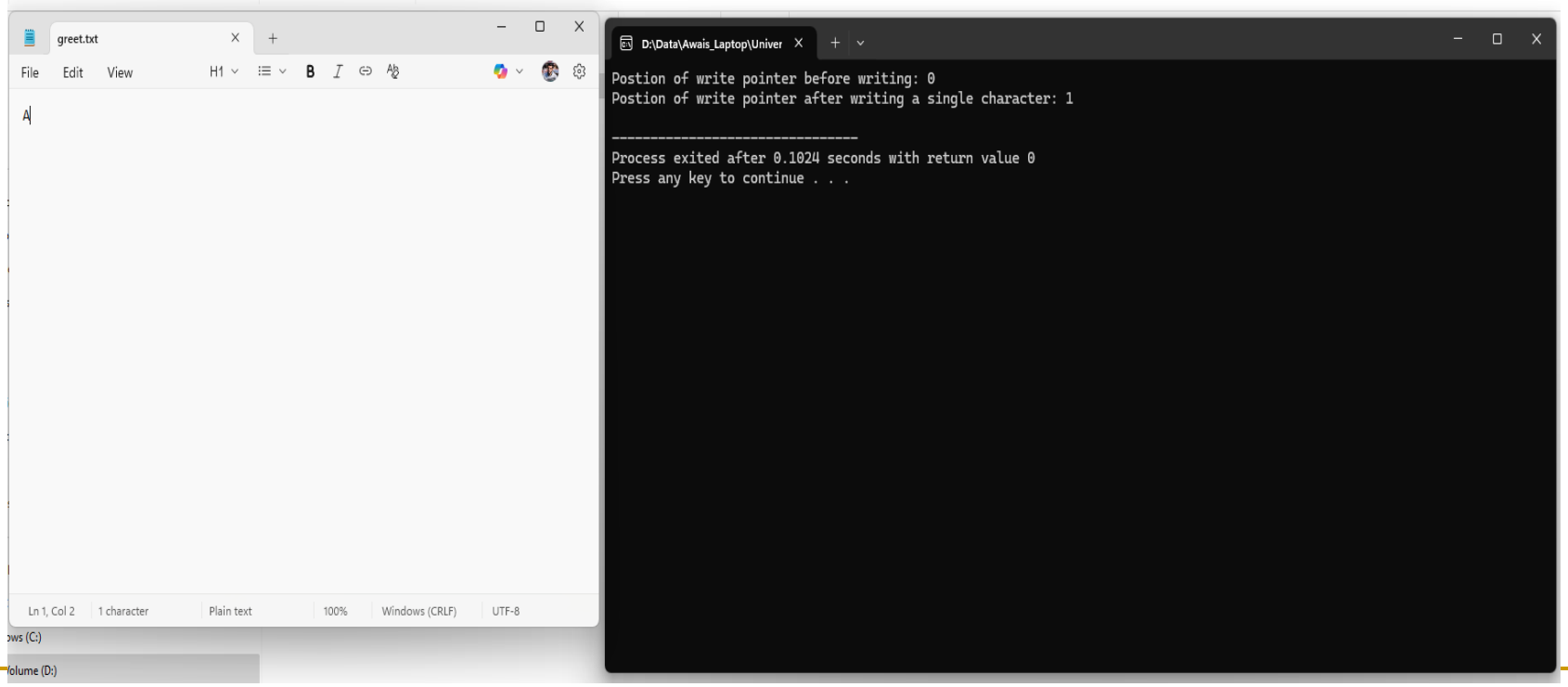
- **tellg()** → Get current get (read) pointer position.
- **tellp()** → Get current put (write) pointer position.
- **seekg(offset, direction)** → Move get pointer.
- **seekp(offset, direction)** → Move put pointer.

Directions:

- ios::beg → Beginning of file.
- ios::cur → Current position.
- ios::end → End of file.

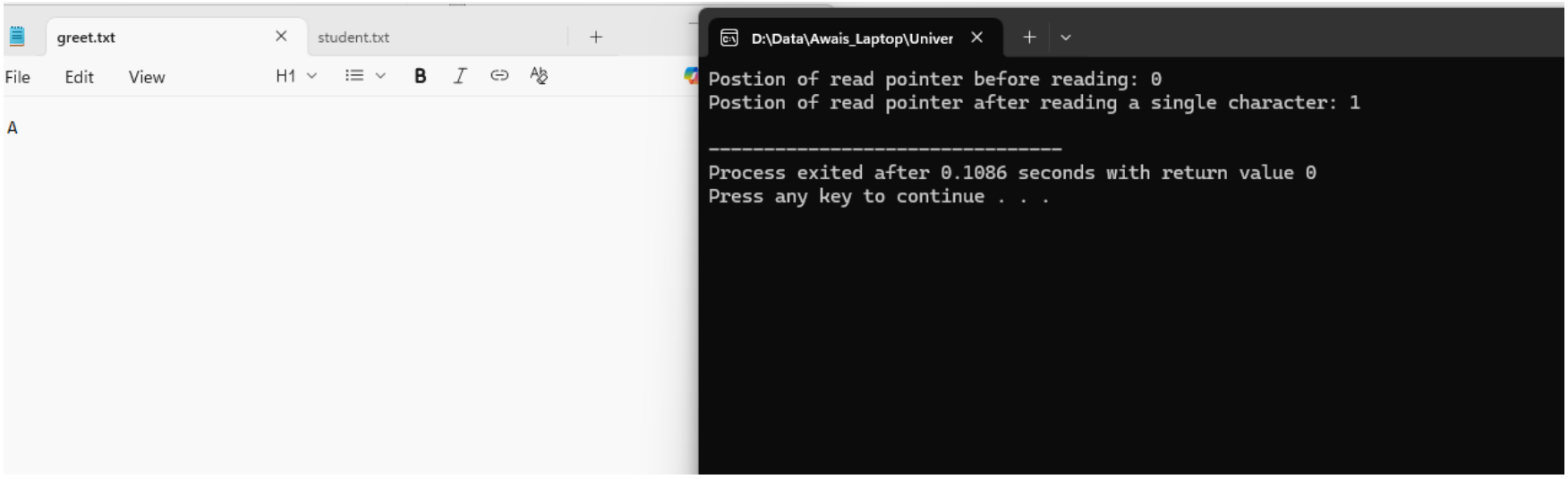
tellp() → Get current put (write) pointer position

```cpp
#include<iostream>
#include<fstream>
using namespace std;
int main(){
    fstream out("greet.txt", ios::out);
    cout << "Postion of write pointer before writing: " << out.tellp() << endl;
    out << "A";
    cout << "Postion of write pointer after writing a single character: " << out.tellp() << endl;
    out.close();
    return 0;
}
```



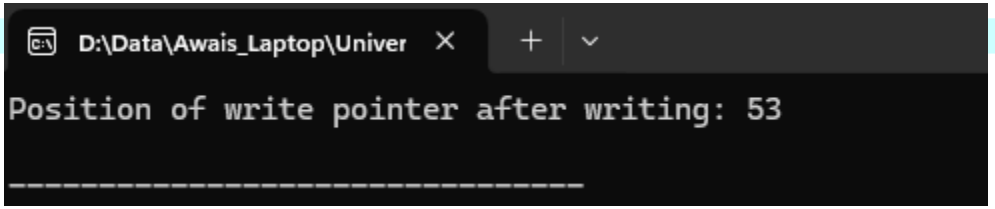
tellg() → Get current get (read) pointer position.

```cpp
#include<iostream>
#include<fstream>
using namespace std;
int main(){
    fstream in("greet.txt", ios::in);
    cout << "Postion of read pointer before reading: " << in.tellg() << endl;
    char ch;
    in.get(ch);
    cout << "Postion of read pointer after reading a single character: " << in.tellg() << endl;
    in.close();
    return 0;
}
```



seekp(offset, direction) → Move put pointer.

```cpp
#include<iostream>
#include<fstream>
using namespace std;
int main(){
    fstream file("student.txt", ios::in | ios::out | ios::trunc);
    file << "01,Awais,3.5 ";
    file << "02,Kashif,3.8 ";
    file << "03,Zahid,2.5 ";
    file << "04,Ahmed,4.0 ";
    cout << "Position of write pointer after writing: " << file.tellp() << endl;
    //one record takes 13 characters and the moment pointer is at 53 means next writing will be start from
    //53 but lets move it back to override the last line
    //53-13 = 40
    //file.seekg(file.tellp()-13); //absoulte postion without direction.
    file.seekp(-13, ios::cur); //relative with reference to current position.
    file << "05,Nadir,2.78";
    file.close();
    return 0;
}
```




```
01,Awais,3.5 02,Kashif,3.8 03,Zahid,2.5 05,Nadir,2.78
```

seekg(offset, direction) → Move get pointer.

```cpp
#include<iostream>
#include<fstream>
using namespace std;
int main(){
    fstream file("student.txt", ios::in | ios::out);
    cout << "Position of pointer before reading: " << file.tellg() << endl;
    char ch;
    cout << "Move the read pointer position to 13" << endl;
    file.seekg(13);
    while(file.get(ch)){
        cout << "Character Read: "<< ch << " Position of pointer after reading: " << file.tellg() << endl;
    }
    file.close();
    return 0;
}
```

```
01,Awais,3.5 02,Kashif,3.8 03,Zahid,2.5 05,Nadir,2.78
```

```
D:\Data\Awais_Laptop\Univer
Position of pointer before reading: 0
Move the read pointer position to 13
Character Read: 0 Position of pointer after reading: 14
Character Read: 2 Position of pointer after reading: 15
Character Read: , Position of pointer after reading: 16
Character Read: K Position of pointer after reading: 17
Character Read: a Position of pointer after reading: 18
Character Read: s Position of pointer after reading: 19
Character Read: h Position of pointer after reading: 20
Character Read: i Position of pointer after reading: 21
Character Read: f Position of pointer after reading: 22
Character Read: , Position of pointer after reading: 23
Character Read: 3 Position of pointer after reading: 24
Character Read: . Position of pointer after reading: 25
Character Read: 8 Position of pointer after reading: 26
Character Read:   Position of pointer after reading: 27
Character Read: 0 Position of pointer after reading: 28
Character Read: 3 Position of pointer after reading: 29
Character Read: , Position of pointer after reading: 30
Character Read: Z Position of pointer after reading: 31
Character Read: a Position of pointer after reading: 32
Character Read: h Position of pointer after reading: 33
Character Read: i Position of pointer after reading: 34
Character Read: d Position of pointer after reading: 35
Character Read: , Position of pointer after reading: 36
Character Read: 2 Position of pointer after reading: 37
Character Read: . Position of pointer after reading: 38
Character Read: 5 Position of pointer after reading: 39
Character Read:   Position of pointer after reading: 40
Character Read: 0 Position of pointer after reading: 41
Character Read: 5 Position of pointer after reading: 42
Character Read: , Position of pointer after reading: 43
Character Read: N Position of pointer after reading: 44
Character Read: a Position of pointer after reading: 45
Character Read: d Position of pointer after reading: 46
Character Read: i Position of pointer after reading: 47
Character Read: r Position of pointer after reading: 48
Character Read: , Position of pointer after reading: 49
Character Read: 2 Position of pointer after reading: 50
Character Read: . Position of pointer after reading: 51
Character Read: 7 Position of pointer after reading: 52
Character Read: 8 Position of pointer after reading: 53

--------------------------------
Process exited after 0.1124 seconds with return value 0
Press any key to continue . . .
```

References

- https://www.geeksforgeeks.org/cpp/file-handling-c-classes/
- https://www.w3schools.com/cpp/ref_fstream_fstream.asp